NYU DBF Mission Profile Version 2

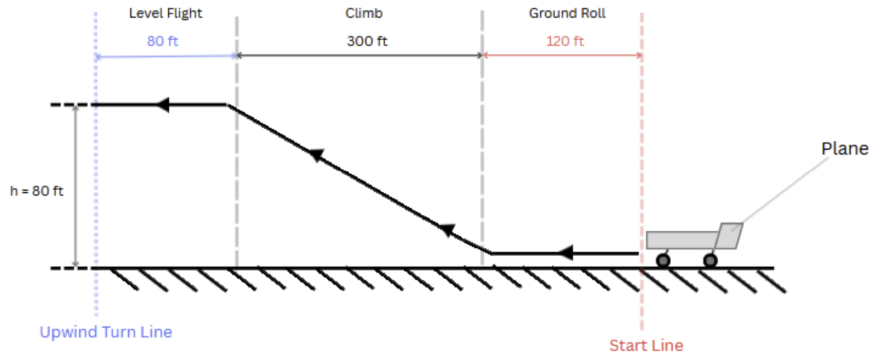


Figure 1: Airplane take-off and climb profile before first 180° turn. Note total takeoff section spans 500 ft.

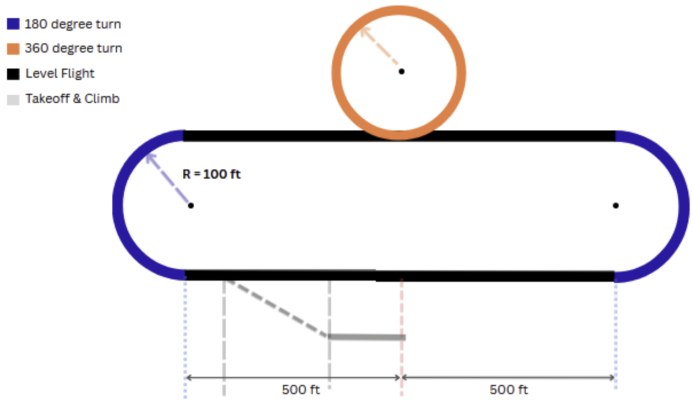


Figure 2: Airplane flight course. Note two 180° turns and single 360° turn. Total straight section spans 1000 ft.

Parameter	**Requirement**
Weight	Maximum take-off weight cannot exceed 18.0 lb, allocated as a 12.0 lb empty weight and 6.0 lb maximum payload.
Cruise	The aircraft should reach a cruise velocity of 80 ft/s.
Takeoff distance	The aircraft takeoff distance is at most 120 feet.
Climb/Altitude	The aircraft must cruise at 80 feet above ground level after 300 ft of climb.
Stall Speed	Stall speed must be less than 48 ft/s at max weight.
Turn Radius	The aircraft should achieve a level-flight, steady turn radius of 100 ft
Battery Capacity	Total mission consumption should leave the plane with a remaining battery capacity of at least 15% of total allotted energy (100 Wh)
Distance	The aircraft must accomplish 5 laps within 5 minutes under maximum weight.

General Configuration Selection

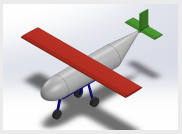
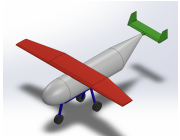
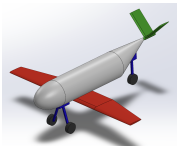
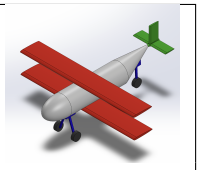
Category	**Weight**	 **Concept 1**	 **Concept 2**	 **Concept 3**	 **Concept 4**
Cruise Aerodynamics	0.10	3	4	5	2
Takeoff/Stall	0.15	3	2	2	5
Manufacturing	0.20	5	2	3	1
Empty Weight	0.10	3	4	5	1
Stability/Handling	0.10	4	4	2	4
Payload Integration	0.10	3	3	5	2
Turning	0.15	4	4	2	3
Ground/Landing	0.10	4	4	1	2
Total	1.00	**3.75**	3.20	3.00	2.50

Table 1: Weighted configuration matrix (1 = poor, 5 = excellent). The selected concept is shaded.

Four candidate configurations were compared using eight weighted metrics, each scored from 1 (poor) to 5 (excellent). Weights were assigned based on the mission requirements. All concepts assume a tractor motor, and propulsion was evaluated separately.

Mid-wing was eliminated early due to harder integration with the fuselage, as well as interference with payload storage. A T-tail layout was eliminated early, as designing against bending moments at the tip of the vertical stabilizer would require more weight at the end of the aircraft, which likely alters the CG position. An elliptical planform was eliminated due to difficulty in manufacturing.

The candidates are as follows:

1) Rectangular planform, high wing, tricycle landing gear, conventional tail
2) Tapered planform, high wing, tricycle landing gear, H-tail
3) Tapered planform, low wing, taildragger landing gear, V-tail
4) Rectangular planform, biplane, taildragger landing gear, conventional tail

Cruise Aerodynamics

Concept 3 scored the highest because its tapered planform reduces induced drag during steady flight compared to the rectangular planform, and its V-tail has fewer junctions for less interference drag and likely a smaller wetted area than the conventional tail and H-tail. Concept 2 benefits from the tapered planform, but its H-tail increases wetted surface area. Concept 1 produces more induced drag, and Concept 4's biplane structurally requires struts between the two wings, creating a significant source of drag.

Battery consumption is included in this evaluation, as more thrust, and therefore more power, is required to maintain steady flight as cruise drag increases. Since cruise makes up the majority of the flight path, battery consumption is largest during cruise.

Takeoff and Stall

Concept 4 scores the highest since the two wings provide a large wing area within the competition limits, and make significantly smaller wingloadings, and stall speeds, feasible. Concept 1's rectangular planform stalls first at the root, so the control surfaces are likely still able to function. Concepts 2 and 3 tend to stall first at the tips due to the tapering wing, and while Concept 3 benefits from ground effect, its taildragger gear presents a risk of ground looping.

Manufacturing

Concept 1 is the simplest to build, with two parallel front and rear spars in its rectangular wing. Concepts 2 and 3 require a modified rib and spar structure due to the tapering wing planform, and both have tails that introduce more complexity than the conventional tail design. Concept 4's biplane configuration demands a more complicated structural design, with two wings attached differently to the fuselage. This metric received the highest weight since there is no previous year's reference; with less certainty about what works and what does not, a safer, simpler design is preferred.

Empty Weight

Concept 3 is the lightest due to its smaller V-tail, taildragger gear, and tapering that brings the center of pressure closer to the root (and hence requires less structure to reinforce the root). Concept 2 has a larger H-tail and the larger tricycle gear configuration, which are likely slightly heavier. Concept 1 is heavier since it has a full rectangular planform, and Concept 4 is the heaviest due to its two wings and the struts required to support the biplane structure.

Stability and Handling

Concepts 1, 2, and 4 all perform better than Concept 3. Namely, high-wing configurations tend to be more laterally stable, while the biplane configuration concentrates wing area near the fuselage, so there is less roll inertia than in its monoplane counterparts. Concept 3 must couple its rudder and elevator functions into two control surfaces, which not only increases control complexity but likely makes the aircraft more difficult for the RC pilot to handle.

Payload Integration

Concept 3 scored highest because the low wing leaves the top of the fuselage open for a payload hatch, which is the easiest to access. Concepts 1 and 2 require side or bottom hatches, while Concept 4 performs the most poorly since the biplane wing configuration crowds the fuselage further.

Turning

Banking the aircraft demands more lift to maintain level flight while turning. In addition, roll capability and drag during the turn are also considered. With the 80 ft/s target speed and the 100 ft target turn radius, the bank angle is estimated at around 63°, so during the turn the aircraft must generate more than twice its cruise lift. Stall behavior is therefore considered. Concepts 1 and 2 score the same due to a tradeoff between drag and root-first versus tip-first stall. Concept 4 ranks behind them: while its roll inertia is low and its structure braces against the centripetal load, the drag of its complex wing structure is a tradeoff. Concept 3 ranks the worst since its coupled tail control surfaces increase the difficulty of completing coordinated turns, combined with the same tip-stall tendencies as Concept 2.

Ground Handling and Landing

The tricycle gear of Concepts 1 and 2 is less prone to ground looping than the taildragger gear of Concepts 3 and 4. In addition, Concept 3's low wing adds a risk of the wingtips striking something on the ground.

Results and Hybrid Configuration

Concept 1 scored highest at 3.75, followed by Concept 2 (3.20), Concept 3 (3.00), and Concept 4 (2.50). Concept 1's lead comes primarily from manufacturing simplicity and low-speed performance, while Concept 2 holds advantages in cruise efficiency and empty weight, both resulting from its tapered planform.

To capture part of that advantage, refining Concept 1's planform with tapered outer panels and a constant-chord center panel was discussed. Although more difficult to manufacture, it would improve expected cruise performance. It was decided to move forward with lightly tapered outer panels, similar to the planform of a Cessna 172. The tricycle landing gear and conventional tail configurations were maintained.

Propeller Configuration Selection

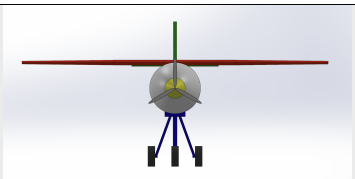
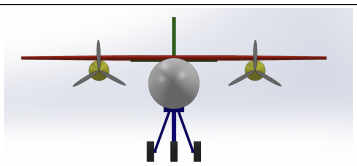
Category	**Weight**	 **Single nose tractor**	 **Dual wing-mounted tractor**
Weight & Manufacturing	0.4	2	1
Efficiency	0.2	2	1
Thrust & Power	0.3	1	2
Drag	0.1	1	2
Total	1.0	**1.6**	1.4

Table 2: Propeller configuration matrix (concepts ranked 1 or 2). The selected configuration is shaded.

Before configuration selection, a tractor layout was chosen over a pusher. A pusher propeller is closest to the ground during takeoff rotation, increasing the risk of a propeller strike or needing to reduce propeller diameter. It also operates in the disturbed wake of the fuselage and wing, reducing propulsive efficiency, and moving the motor mass aft shifts the center of gravity toward the rear.

Since this aircraft is intended as a common baseline for future iterations, the tractor layout was also preferred for its extensive base of design references. All configuration candidates were therefore evaluated with a tractor propeller; the choice between single and twin tractors is addressed here. Note that each configuration has its own aerodynamic drawbacks, so more weight was placed on thrust and manufacturing.

The concepts are ranked 1 or 2. The two candidates are:

1) Single nose tractor
2) Dual wing-mounted tractors

Weight and Manufacturing

The single nose tractor requires less structural support since its mount can be integrated directly into the fuselage, while the dual wing-mounted tractors require a dedicated mounting design on the wings, which demands more extensive design of the wing's internal rib and spar structure.

Efficiency

A single larger propeller is more efficient than a smaller propeller on each of the dual-mounted motors. For the thrust required by the aircraft, a larger propeller on the nose accelerates a larger mass of air through a smaller velocity increment, which yields a higher propulsive efficiency than two smaller propellers on the wings [2].

Thrust and Power

Two motors combined have the capacity to impart more total power into the incoming flow, and therefore generate more thrust, than one motor alone.

Drag

Both configurations produce drag. However, the single nose propeller energizes the flow next to the fuselage all the way to the tail, so the turbulent skin friction across this surface area is likely larger than the added drag near the wing surfaces.

Results

The single nose tractor scores higher than the dual wing-mounted tractor configuration, 1.6 to 1.4. It leads primarily due to easier manufacturing and reduced weight, despite its lower total thrust.

Final Conceptual Design

The final conceptual aircraft is illustrated in Fig. 3. The aircraft is a high-wing monoplane with a conventional tail, single-tractor motor, tricycle landing gear, and hybrid tapered planform.

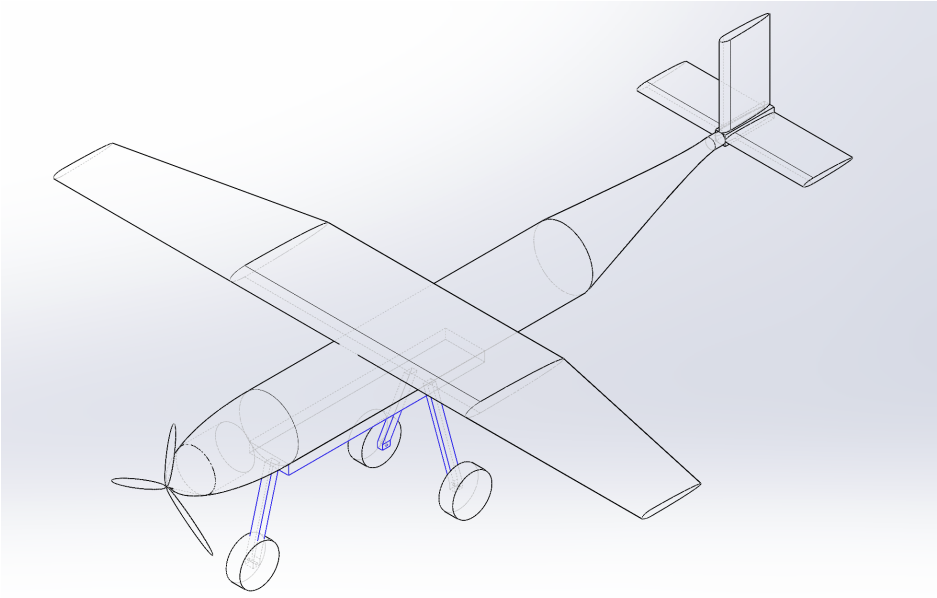


Figure 3: Final Conceptual Design

Constraint Analysis

From Gudmundsson's *General Aviation Aircraft Design: Applied Methods and Procedures* [1], the following thrust-to-weight ratio and wing loading relationships were used to select a design point for the aircraft: target ground roll, target cruise speed, target constant-velocity turn, and target climb angle. These values are set directly from the mission profile.

Note that k is the induced drag factor from Prandtl's lifting line theory and q is the dynamic pressure of the incoming airstream. n is the load factor during turning, where ϕ is the bank angle. Note that $e = (1+\delta)^{-1}$, where $e = 1$ for an elliptical lift distribution. For constraint analysis, e is estimated to be 0.7. Note that $\rho = 0.002372\ slug/ft^3$ for 80 ft above sea level.

$$k = \frac{1}{\pi e AR}, \qquad q = \frac{1}{2}\rho V^2, \qquad n = \frac{1}{\cos\phi}, \qquad \phi = \tan^{-1}\left(\frac{V^2/R}{g}\right) \tag{1}$$

Stall Speed

First, a hard upper wing loading bound is given from the target maximum stall speed. $C_{L,max}$ is estimated to be 1.3 from reading up on previous DBF aircraft.

$$\left(\frac{W}{S}\right)_{max} = \frac{1}{2}\rho V_{stall}^2 C_{L,max} \tag{2}$$

Takeoff Ground Roll

From the 120 ft target ground roll, the following constraint presented by Gudmundsson is used [1, Eq. (3-1)]. Note that the takeoff drag and lift coefficients are adapted from previous reports that matched the selected aircraft configuration ($C_{D,TO} = 0.07$; $C_{L,TO} = 0.7$), and S_G is the ground roll distance.

$$\frac{T}{W} = \frac{1.21}{g\rho C_{L,max} S_G}\left(\frac{W}{S}\right) + \frac{0.605}{C_{L,max}}\left(C_{D,TO} - \mu C_{L,TO}\right) + \mu \tag{3}$$

Cruise

From the 80 ft/s target cruise speed, the following constraint is used [1, Eq. (3-10)]. Again, note that $C_{D,min} = 0.04$ is estimated from previous trends.

$$\frac{T}{W} = qC_{D,min}\left(\frac{1}{W/S}\right) + \frac{k}{q}\left(\frac{W}{S}\right) \tag{4}$$

Climb

From the 80 ft altitude and 300 ft climb distance, the climb angle $\gamma = \tan^{-1}(80/300)$ is calculated. The following constraint is given [1, Eq. (3-6)]; note that it is independent of wing loading.

$$\frac{T}{W} = \sin\gamma + \sqrt{4kC_{D,min}} \tag{5}$$

Level Constant-Velocity Turn

The fourth constraint relationship comes from the target 100 ft turn radius, specifically for a constant-velocity level turn, where the velocity is the target cruise speed [1, Eq. (3-7)].

$$\frac{T}{W} = q\left[\frac{C_{D,min}}{W/S} + k\left(\frac{n}{q}\right)^2\left(\frac{W}{S}\right)\right] \tag{6}$$

Design Point Selection

The methodology in using these equations is to select a design point, with reasonable margin, from the optimal thrust-to-weight ratio and wing loading combination. Aspect ratio (AR) plays a key role in determining whether a design point is feasible within the 6 ft maximum wingspan set by the competition. Recall that the MTOW is 18 lb. The following relations are important:

$$b = (S \times AR)^{1/2}, \qquad c = \frac{b}{AR} \tag{7}$$

Iteratively, $AR = 6$ was tried, from which a design point of $W/S = 3.1$ and $T/W = 0.4$ was selected. This evaluates to a wingspan of 5.9 ft, which leaves a margin of roughly half an inch from the competition limit on each wing. While possible, this was judged too close to the upper limit, so $AR = 5$ was tried next.

For $AR = 5$, the constraint relations were recalculated, and the design point of $W/S = 3.1$ and $T/W = 0.4$ was selected, which is at least a 5% margin above the required T/W dictated by the climb angle constraint (Fig. 4). This evaluates to a wingspan of 5.38 ft, rounded up to the next inch at 65 inches, and a mean chord of 12.9 inches, rounded up to 13 inches. This was more workable and was set as the target design point.

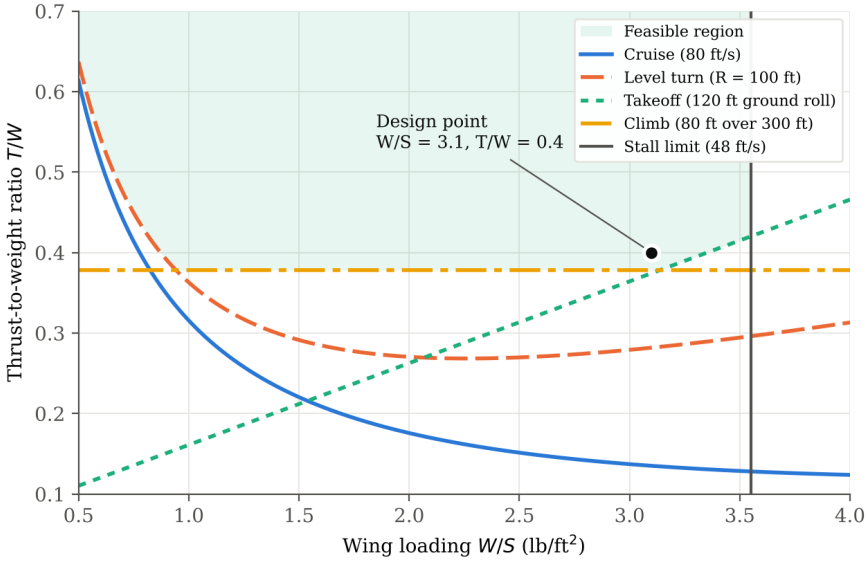


Figure 4: Constraint diagram for $AR = 5$. The shaded area satisfies every constraint; the selected design point lies above the climb and takeoff lines and to the left of the stall limit.

The following sizing was selected:

Parameter	**Value**
Wing loading, W/S	3.1 lb/ft^2
Thrust-to-weight ratio, T/W	0.4
Aspect ratio, AR	5.0
Wingspan, b	65 in
Mean chord, c_{avg}	13 in

Table 3: Selected baseline sizing.

References

[1] S. Gudmundsson, *General Aviation Aircraft Design: Applied Methods and Procedures*, 2nd ed. Oxford: Butterworth-Heinemann (Elsevier), 2022. ISBN 978-0-12-818465-3.

[2] Massachusetts Institute of Technology, "Thermodynamics and Propulsion," 16.Unified Engineering course notes. [Online]. Available: `https://web.mit.edu/16.unified/www/FALL/thermodynamics/notes/node86.html`

Figures 1,2 created via Canva
Concept Figures and Figure 3 from SOLIDWORKS
Figure 4 created via Python matplotlib